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Part A

Q1. Data is raw, unprocessed facts, figures, or observations about something until they are organised or analysed - on their own they don't carry any meaning.

Example not in the dataset: the number of steps a person takes per day.

Q2. Structured - the data is presented in a table with an established structure of rows (students) and columns (variables), and each column has a consistent data type.

All variables are uniform in format - Study_Hours is always a number, Programme is always one of a limited number of category labels. This is typical of structured, relational, table-oriented data like a spreadsheet or a database table.

Q3. Quantitative data is numeric, measures a quantity or amount, and can be used in mathematical calculations. Example: Study_Hours.

Qualitative data describes a quality, label or characteristic and is not numerical. Example: Programme (IT or Business).

Q4.

Student_ID - Qualitative. It is an identifier, not a measurable quantity, so despite looking numeric it cannot be meaningfully averaged.

Programme - Qualitative. A category label (IT or Business).

Study_Hours - Quantitative. An integer number of hours.

Sleep_Hours - Quantitative. A number representing time in hours.

Satisfaction_Level - Qualitative. A number (1-5) that represents a level of feeling rather than a measured quantity, so easy to mistake for quantitative.

Attendance_Percentage - Quantitative. A numeric measurement.

Test_Score - Quantitative. A numeric measurement.

Q5.

Student_ID - Nominal. Labels only, with no order or numeric meaning.

Programme - Nominal. Categories with no natural ordering.

Study_Hours - Ratio. Has a true zero and equal, meaningful intervals.

Sleep_Hours - Ratio. A true zero point with equal intervals between hours.

Satisfaction_Level - Ordinal. Has a definite sequence, but the difference between levels is not guaranteed to be equal.

Attendance_Percentage - Ratio. True zero, equal intervals as a percentage.

Test_Score - Ratio. Equal intervals and a true zero.

Q6. Student_ID is a nominal variable - the digits carry no quantitative meaning, so it would be meaningless to calculate an average of them; they are simply used to identify a student.

Satisfaction_Level is still orderable even though the psychological distance between, say, 3 and 4 is not guaranteed to equal the distance between 4 and 5, because it has a clear underlying order - 5 is more satisfied than 4, which is more than 3 - even without a confirmed equal interval.

Screenshots

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C:\Users\Shivaar\AppData\Local\Programs\Python\Python313\python.exe "C:\Users\Shivaar\PycharmProjects\DataScience500_course\DS500_Practical2_402308101\practical2.py"
[72 85 64 91 75 82 60 88 79 55 70 84] int64
[88 92 76 95 84 89 73 94 86 68 81 90] int64
[ 6 10  4 12  7  9  5 11  8  3  6 10] int64
75.42 77.0
55 91 36
10.99
[85 91 82 88 79 84] 6
[88 92 76 95]
[ 3  6 10]
[72 85 64 91 75 82 60 88 79 55 70 84]
[77 90 69 96 80 87 65 93 84 60 75 89]
  Student_ID Programme  ... Attendance_Percentage Test_Score
0      S001      IT  ...                88          72
1      S002      IT  ...                92          85
2      S003 Business  ...                76          64
3      S004      IT  ...                95          91
4      S005 Business  ...                84          75
5      S006      IT  ...                89          82
6      S007 Business  ...                73          60
7      S008      IT  ...                94          88
8      S009 Business  ...                86          79
9      S010      IT  ...                68          55
10     S011 Business  ...                81          70
11     S012      IT  ...                90          84

[12 rows x 7 columns]
  Student_ID Programme  ... Attendance_Percentage Test_Score
0      S001      IT  ...                88          72
1      S002      IT  ...                92          85
2      S003 Business  ...                76          64
3      S004      IT  ...                95          91
4      S005 Business  ...                84          75

```

```

[5 rows x 7 columns]
(12, 7)
Index(['Student_ID', 'Programme', 'Study_Hours', 'Sleep_Hours',
      'Satisfaction_Level', 'Attendance_Percentage', 'Test_Score'],
      dtype='str')
Student_ID      str
Programme       str
Study_Hours     int64
Sleep_Hours     float64
Satisfaction_Level int64
Attendance_Percentage int64
Test_Score      int64
dtype: object
  Student_ID Programme Test_Score
0      S001      IT      72
1      S002      IT      85
2      S003 Business    64
3      S004      IT      91
4      S005 Business    75
5      S006      IT      82
6      S007 Business    60
7      S008      IT      88
8      S009 Business    79
9      S010      IT      55
10     S011 Business    70
11     S012      IT      84
Student_ID      S004
Programme       IT
Study_Hours      12
Sleep_Hours      5.5
Satisfaction_Level 2
Attendance_Percentage 95
Test_Score       91
Name: 3, dtype: object

```

```
Attendance_Percentage    95
Test_Score                91
Name: 3, dtype: object

   Student_ID  Programme  Test_Score  Attendance_Percentage
0         S001         IT          72                   88
1         S002         IT          85                   92
3         S004         IT          91                   95
4         S005  Business          75                   84
5         S006         IT          82                   89
7         S008         IT          88                   94
8         S009  Business          79                   86
10        S011  Business          70                   81
11        S012         IT          84                   90

Programme
Business    69.60
IT          79.57
Name: Test_Score, dtype: float64

   Student_ID  Test_Score  Performance
0         S001          72         Pass
1         S002          85         Pass
2         S003          64         Pass
3         S004          91         Pass
4         S005          75         Pass
5         S006          82         Pass
6         S007          60         Pass
7         S008          88         Pass
8         S009          79         Pass
9         S010          55         Pass
10        S011          70         Pass
11        S012          84         Pass

Process finished with exit code 0
```

Part C

Q20. The mean Test_Score (75.42) is slightly lower than the median (77.00), indicating that a few lower scores (55, 60) are dragging the mean down compared to the typical student.

Range: 55 to 91, a spread of 36 marks, with a population standard deviation of 10.99, showing a moderate spread across the class.

The mean Test_Score for IT students (79.57) is higher than for Business students (69.60), and a Test_Score above 70 combined with attendance above 80% corresponds mostly to IT students.